NHS Antidepressant Prescribing Analysis

End-to-End Data Analytics Portfolio Project

England · January 2021 – December 2025

445,116,238 items · £1,204,000,000 total cost · 32 drugs · 7 NHS regions · 60 months

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Role: Data Analyst

Data Source: NHS Business Services Authority · Prescription Cost Analysis (PCA)

Licence: Open Government Licence v3.0

Year: 2025

1. Executive Summary

This project delivers a comprehensive end-to-end data analysis of NHS antidepressant prescribing patterns across England from January 2021 to December 2025, covering 60 months of monthly prescription data sourced directly from the NHS Business Services Authority (NHS BSA) Prescription Cost Analysis (PCA) dataset.

The analysis examines prescribing volume, cost trends, drug-level efficiency, and regional variations across all seven NHS England regions, culminating in a 12-month forward forecast using the Facebook Prophet time series model.

Key Finding

NHS antidepressant prescribing volumes rose consistently from 83M items in 2021 to 95M in 2025 (+14.6%), while total costs fell from £291M to £224M (-23.1%) over the same period. This divergence is driven primarily by the genericisation of Sertraline hydrochloride in early 2022, which reduced its mean cost per item by 37.6% in a single year.

Key Headline Metrics

Metric	Value	Period
Total Items Prescribed	**445,116,238**	Jan 2021 – Dec 2025

Total Prescribing Cost	**£1,204,000,000**	Jan 2021 – Dec 2025
Mean Cost Per Item	**£2.70**	Overall average
Most Prescribed Drug	**Sertraline hydrochloride**	115,001,208 items (25.8%)
Highest Cost Per Item Drug	**Tranylcypromine sulfate**	£1,337.24 per item
Fastest Growing Region	**North East And Yorkshire**	+26.3% items 2021–2025
Highest Cost Per Item Region	**South East**	£3.31 per item

2. Project Objectives

This project was designed to answer seven analytical questions that reflect the priorities of NHS commissioners and data analysts working in primary care analytics.

1. What are the national trends in antidepressant prescribing volume and cost over the period 2021–2025?
2. What drove the sharp reduction in prescribing costs observed in 2022?
3. Which drugs represent the greatest cost burden relative to their prescribing volume?
4. How do prescribing patterns and costs vary across NHS England regions?
5. Which regions are experiencing the fastest growth in antidepressant demand?
6. Are there seasonal patterns in monthly prescribing volumes?
7. What are the projected prescribing volumes and costs for the next 12 months?

3. Data Source

3.1 NHS BSA Prescription Cost Analysis (PCA)

Attribute	Details
Source	**NHS Business Services Authority (NHS BSA)**
Dataset	Prescription Cost Analysis (PCA) Monthly Data
Coverage	**January 2021 – December 2025 (60 months)**
Geography	7 NHS England Regions
Raw Granularity	GP Practice level – one row per drug per practice per month
Processed Granularity	Aggregated to drug-region-month level (12,328 rows)
Drugs Covered	32 BNF Chapter 4.3 antidepressant chemical substances

Licence	Open Government Licence v3.0

3.2 Key Variables

- YEAR_MONTH – Period of prescribing in YYYY-MM format
- REGION_NAME – NHS England region name (standardised to Title Case)
- BNF_CHEMICAL_SUBSTANCE – Drug chemical substance name from the British National Formulary
- ITEMS – Number of prescription items dispensed
- NIC – Net Ingredient Cost in GBP – the cost of the drug before any discounts

4. Technical Architecture

4.1 Technology Stack

Layer	Technology	Purpose
Ingestion	**Python · requests**	Automated monthly scraping from NHS BSA portal with incremental loading
Processing	**Python · pandas**	11-step data cleaning, filtering and aggregation pipeline
Forecasting	**Python · prophet**	12-month Prophet forecast with 80% confidence intervals and cross-validation
Storage	**MySQL 8.0**	Star schema database – dates, regions, drugs, prescriptions, forecast
Analysis	**SQL · Jupyter**	20 SQL queries across 6 analytical sections and exploratory notebook
Visualisation	**Power BI Desktop**	5-page interactive dashboard with 70+ DAX measures across 8 display folders

4.2 Data Pipeline

The pipeline follows the Extract-Transform-Load (ETL) pattern with each stage implemented as a standalone Python script.

scraper.py – Data Ingestion

- Scrapes monthly PCA CSV files directly from the NHS BSA portal
- Implements a raw landing zone (pca_data/raw/) for data lineage
- Maintains a download log for incremental loading – skips already-downloaded files

processor.py – Data Processing (11 Steps)

8. Load combined raw CSV from landing zone
9. Drop rows with missing values in key fields
10. Strip whitespace from text columns to prevent silent duplicates
11. Filter to 32 antidepressant BNF chemical substances only
12. Standardise region names to Title Case
13. Aggregate from GP practice level to drug-region-month level using groupby().agg()
14. Derive YEAR column from YEAR_MONTH before format conversion
15. Convert YEAR_MONTH from YYYYMM integer to YYYY-MM ISO 8601 string
16. Enforce correct data types (YEAR: int, ITEMS: int, NIC: float)
17. Select and order final columns
18. Save staged output to pca_data/staged_pca_data.csv

forecast.py – Forecasting

- Reads staged_pca_data.csv and aggregates to monthly national totals
- Fits three separate Facebook Prophet models: Items, NIC, and Cost Per Item
- Adds an explicit changepoint at January 2022 to capture the Sertraline genericisation structural break
- Runs walk-forward cross-validation and reports MAPE per model
- Generates 12 months of forward projections with 80% confidence intervals
- Saves combined historical and forecast table to pca_data/forecast.csv

4.3 Database Schema

Table	Type	Description
dates	Dimension	One row per month – date_id, year_month, year, month, month_name
regions	Dimension	One row per NHS region – region_id, region_name
drugs	Dimension	One row per drug – drug_id, bnf_chemical_substance
prescriptions	Fact	One row per drug-region-month – items prescribed and NIC cost with FK references
forecast	Forecast	One row per month – historical actuals + 12-month Prophet forecast with 80% CI

5. Key Findings

5.1 National Prescribing Trends

NHS antidepressant prescribing volumes grew consistently throughout the analysis period, rising from 83M items in 2021 to 95M items in 2025. Despite this growth, total prescribing costs fell from £291M to £224M – a 23.1% reduction over four years.

Year	Items	Total Cost	Cost Per Item	YoY Cost Change
2021	83M	£291M	£3.51	–
2022	86M	£226M	£2.63	**-22.3%**
2023	89M	£232M	£2.61	+2.7%
2024	92M	£230M	£2.50	-0.9%
2025	95M	£224M	£2.36	-2.6%

5.2 The Genericisation Effect – Sertraline Hydrochloride

The single most significant finding is the sharp cost reduction in early 2022, caused by the widespread availability of generic Sertraline hydrochloride. The cost per item for Sertraline fell from £3.54 in 2021 to £2.21 in 2022 – a 37.6% reduction in a single year. Sertraline's £23M annual saving accounts for the majority of the £65M national cost reduction in 2022.

Sertraline is the most prescribed antidepressant in England with 115,001,208 items (25.836% of all prescriptions) but only 21.899% of total cost – a 3.94 percentage point efficiency gap that exists entirely because of genericisation.

5.3 Drug Cost Efficiency Analysis

The efficiency gap measures how disproportionate a drug's cost share is relative to its prescribing volume. A positive gap means the drug costs more than its volume justifies (cost burden). A negative gap means it is cost efficient. The threshold used is ±2 percentage points.

Drug	% Items	% Cost	Gap (pp)	CPI	Classification
Sertraline hydrochloride	25.836%	21.899%	**-3.94**	£2.29	Cost Efficient
Venlafaxine	6.476%	17.448%	**+10.97**	£7.29	Cost Burden
Amitriptyline hydrochloride	17.625%	9.569%	**-8.06**	£1.47	Cost Efficient
Mirtazapine	13.973%	6.590%	**-7.38**	£1.28	Cost Efficient
Fluoxetine hydrochloride	8.324%	8.076%	-0.25	£2.62	Neutral
Duloxetine hydrochloride	4.867%	7.095%	**+2.23**	£3.94	Cost Burden
Vortioxetine	0.385%	3.408%	**+3.02**	£23.95	Cost Burden

Trimipramine maleate	0.019%	2.351%	**+2.33**	£342.16	Cost Burden
Tranylcypromine sulfate	0.005%	2.241%	**+2.24**	£1,337.24	Cost Burden

5.4 Regional Analysis

All seven NHS England regions experienced consistent growth in antidepressant prescribing between 2021 and 2025. North East And Yorkshire demonstrated the highest growth at +26.3%, nearly double the national average of +14.6%. South West was the slowest growing region at +4.2%, less than a third of the national average.

Region	Items 2021	Items 2025	Change	Cost 2021	Cost 2025	Cost Change
North East And Yorkshire	17,750,721	22,419,790	**+26.3%**	£54,887,870	£45,126,663	**-17.8%**
Midlands	15,436,017	18,012,360	**+16.7%**	£54,660,729	£43,707,159	**-20.0%**
North West	12,468,335	14,020,170	**+12.4%**	£42,449,534	£31,249,553	**-26.4%**
South East	10,639,433	11,607,354	+9.1%	£44,718,283	£33,548,520	**-25.0%**
East Of England	9,699,149	10,752,215	+10.9%	£32,898,810	£26,161,565	**-20.5%**
London	8,644,409	9,605,475	+11.1%	£32,040,318	£23,029,904	**-28.1%**
South West	8,323,915	8,673,972	+4.2%	£29,824,347	£21,476,388	**-28.0%**

South East recorded the highest mean cost per item at £3.31 compared to North East And Yorkshire with the lowest at £2.33 – a 42% disparity for the same drug class. London and South West achieved the joint-largest cost reductions at -28.1% and -28.0% respectively.

5.5 Seasonal Patterns

The matrix heatmap of monthly prescribing volumes (see images/matrix_heatmap.png) reveals that year-on-year growth is the dominant signal. Every month in 2025 prescribed more items than the same month in 2021 without exception.

February is the only consistent seasonal pattern – it is the lowest or near-lowest prescribing month in all five years, ranging from 6,363,902 items in 2021 to 7,268,417 in 2025. There is no single consistent peak month: December led in 2021 (7,389,273) and 2025 (8,447,750 – the highest single month in the dataset), March led in 2022 and 2023, and October led in 2024 (8,045,117).

The May–June 2024 cost anomaly: a sharp cost spike occurred in May–June 2024 not accompanied by a corresponding volume increase, suggesting a temporary procurement or pricing disruption. This anomaly was treated as an outlier in the forecasting model.

5.6 Key Influencer Analysis

Using Power BI's Key Influencers visual set to Increase, the analysis identifies the three drugs most responsible for pulling the average cost per item upward:

- When bnf_chemical_substance is Tranylcypromine sulfate – average CPI increases by £1,310
- When bnf_chemical_substance is Isocarboxazid – average CPI increases by £356.10
- When bnf_chemical_substance is Trimipramine maleate – average CPI increases by £230.40

These three drugs collectively account for less than 0.03% of total prescriptions yet are the primary statistical drivers of high cost per item. Their extreme unit prices (Tranylcypromine at £1,337.24/item, Isocarboxazid at £349.57/item, Trimipramine at £342.16/item) disproportionately inflate the national average CPI of £2.70, confirming that the mean cost per item is skewed upward by a small number of high-cost specialist drugs.

6. Power BI Dashboard

6.1 Dashboard Overview

Page	Title	Analytical Questions Answered
1	National Overview	National trends, cost per item annual trend, monthly dual-axis line chart
2	Drug Analysis	Drug cost efficiency, top 10 by items/cost/CPI, efficiency classification table
3	Regional Analysis	Scatter plot, annual bar charts by region, performance summary 2021 vs 2025
4	Drivers & Trends	Stacked area charts by drug and region for items/cost/CPI, monthly MoM charts
5	Forecast	12-month Prophet forecast with 80% CI bands, forecast table, KPI cards with YoY labels

6.2 DAX Measures

The data model contains 70+ DAX measures organised across 8 display folders.

Folder	Count	Key Examples
Volume Metrics	8	Total Items, % of Total Items, Sertraline % of Items
Cost Metrics	12	Total Cost, Cost Per Item, Efficiency Gap, Efficiency Classification
Time Intelligence	18	Items YoY %, Cost YoY %, Latest Month Items MoM Change, Items 3M Rolling Avg
Current Year	3	Current Year Items, Current Year Cost, Current Year Cost Per Item
Previous Year	3	Previous Year Items, Previous Year Cost, Previous Year Cost Per Item
Drug Insights	9	Most Prescribed Drug, Drug with Most Cost, Drug with Highest Cost Per Item
Regional Insights	14	Highest Prescribing Region, Fastest Growing Region, Items Change 2021 vs 2025
Forecast	15	Total Forecast Items, Forecast CPI, Forecast Items vs Current Year Label

7. Forecasting Methodology

7.1 Model Configuration

Parameter	Value	Rationale
Training data	**60 months**	January 2021 to December 2025
Forecast horizon	**12 months**	Full calendar year of forward projections
Confidence interval	**80%**	Appropriate for operational planning
Explicit changepoint	**Jan 2022**	Marks the Sertraline genericisation structural break explicitly
changepoint_prior_scale (Items)	0.05	Conservative – items trend is smooth and steady
changepoint_prior_scale (Cost/CPI)	0.30	Higher flexibility – cost had a sharp structural break in 2022
Seasonality	Annual	Weekly/daily seasonality disabled – data is monthly

7.2 Model Validation

Model	MAPE	Interpretation
Items Forecast	**3.12%**	Good accuracy
Cost (NIC) Forecast	13.03%	Moderate – structural break impact (see note)
Cost Per Item Forecast	12.27%	Moderate – structural break impact (see note)

Note on Cost and CPI MAPE: The 13.03% and 12.27% MAPE scores reflect the challenge of cross-validating a model containing a sharp structural break in early 2022. Training folds that precede the 2022 changepoint naturally produce higher errors when forecasting post-2022 values. The final model with the explicit changepoint produces materially better forecasts than these cross-validation scores suggest. The Items MAPE of 3.12% confirms strong model performance on the primary volume measure.

7.3 Forecast Results (January – December 2026)

Month	Items (M)	Lower	Upper	Cost (£M)	Lower	Upper	CPI (£)
2026-01	8.19	8.00	8.35	£19.34	£17.47	£21.15	£2.37
2026-02	7.63	7.46	7.82	£17.99	£16.29	£19.70	£2.40
2026-03	8.38	8.21	8.55	£20.21	£18.45	£22.10	£2.40
2026-04	7.92	7.75	8.09	£17.90	£15.99	£19.96	£2.26
2026-05	8.20	8.02	8.37	£18.68	£16.67	£20.80	£2.27
2026-06	8.21	8.06	8.39	£18.98	£17.08	£21.15	£2.29
2026-07	8.24	8.08	8.42	£19.37	£17.28	£21.75	£2.35
2026-08	8.21	8.04	8.38	£19.26	£17.27	£21.54	£2.34
2026-09	8.25	8.08	8.42	£19.75	£17.47	£22.10	£2.40
2026-10	8.29	8.11	8.46	£18.92	£16.80	£21.15	£2.28
2026-11	8.29	8.12	8.46	£18.98	£16.66	£21.15	£2.28
2026-12	8.42	8.25	8.60	£19.89	£17.52	£22.10	£2.36
TOTAL / AVG	**98.24**			**£229.29**			**£2.33**

80% confidence interval · Facebook Prophet · Explicit January 2022 changepoint · Walk-forward cross-validation (5 folds)

8. SQL Analysis

The file analysis.sql contains 20 SQL queries across six analytical sections that translate all key notebook findings into pure SQL against the nhs_prescribing star schema.

Section	Title	Queries Included
1	National Overview	Total items and cost, annual summary, mean monthly cost, min/max distribution
2	Monthly Trends	Full monthly time series, year-over-year cost change
3	Drug-Level Analysis	Top 10 by items, Top 10 by cost, percentage contribution (window functions), annual trends
4	Regional Analysis	Annual items and cost by region, percentage change 2021 vs 2025, monthly regional time series
5	Sertraline Deep Dive	Overall summary, annual trend, monthly trend, cost per item by region, vs national average
6	Forecast Queries	Full forecast table, forward forecast only, 12-month summary totals, actual vs fitted accuracy

9. Recommendations

9.1 Drug Prescribing

1. Review Venlafaxine prescribing pathways

At £7.29/item vs national average of £2.70, and with a +10.97pp cost burden gap, Venlafaxine represents the single largest cost efficiency opportunity. A 10% shift to equivalent generics could save ~£21M annually.

2. Monitor high-unit-cost MAOIs

Tranylcypromine sulfate (£1,337.24/item), Nefazodone (£617.55/item) and Isocarboxazid (£349.57/item) are extreme outliers. Even small volume reductions have disproportionate cost impact.

3. Protect generic Sertraline gains

The 37.6% cost-per-item reduction through genericisation should be preserved. Any formulary changes shifting patients back to branded formulations would reverse £23M+ in annual savings.

9.2 Regional Commissioning

4. Investigate the South East cost premium

The 42% cost-per-item gap between South East (£3.31) and North East And Yorkshire (£2.33) is significant. Aligning South East's drug mix with the more cost-efficient North East And Yorkshire prescribing patterns could yield £8–12M in annual savings.

5. Prepare capacity for North East And Yorkshire

At +26.3% growth over four years, nearly double the national average, this region requires forward planning for GP capacity, formulary management and pharmacy supply chains.

9.3 Demand Planning

6. Budget for continued volume growth

The forecast projects 98.24M items in 2026 (▲3.3% on 2025). NHS commissioners should plan for continued prescribing growth through 2026.

7. Anticipate modest cost increase

Despite continued CPI decline (£2.33 forecast), total costs are projected to rise to £229.29M in 2026 (▲2.2%) purely due to volume growth. Genericisation savings are now largely locked in and unlikely to generate further large cost reductions absent new patent expiries.

10. Project File Structure

File / Folder	Description
scraper.py	Automated NHS BSA data scraper
processor.py	11-step data processing pipeline
forecast.py	Facebook Prophet forecasting – generates forecast.csv
sql/schema.sql	Star schema DDL – creates nhs_prescribing database and all tables
sql/analysis.sql	20 analytical SQL queries across 6 sections
sql/forecast.sql	Forecast table DDL
pca_data/raw/	Landing zone for downloaded CSV files
pca_data/staged_pca_data.csv	Processed data (output of processor.py)
pca_data/forecast.csv	Forecast output (output of forecast.py)
pca_data/logs/	Pipeline execution logs

images/	Dashboard screenshots for README and documentation
Antidepressant_Analysis.ipynb	Exploratory data analysis notebook
nhs_analysis.pbix	Power BI dashboard (5 pages, 70+ DAX measures)
docs/	Project documentation folder
docs/nhs_project_documentation.pdf	Full project documentation – methodology, findings, forecasting and recommendations
requirements.txt	Python dependencies
.env	Environment variable template

11. How to Run This Project

Prerequisites

- Python 3.10+
- MySQL 8.0
- Power BI Desktop

Step-by-Step Instructions

19. Install dependencies: pip install -r requirements.txt
20. Configure .env with your MySQL credentials (copy from .env.example)
21. Run schema.sql and forecast.sql in MySQL Workbench to create the database
22. Scrape the data: python scraper.py
23. Process the data: python processor.py
24. Generate the forecast: python forecast.py
25. Open nhs_analysis.pbix in Power BI Desktop and refresh all data sources

12. Author

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Project Type	Portfolio Project – NHS Primary Care Analytics
Data Source	NHS Business Services Authority – Open Government Licence v3.0
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Licence v3.0